JOSEPH SRUJAN EDA

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PROFESSIONAL SUMMARY

Embedded Software Engineer with ~3+ years of experience in **Embedded Linux**, **BSP Development**, **Device Drivers**, and **RTOS/Bare-Metal Programming**. Proven track record in board bring-up, Linux kernel tuning, Yocto layer customization, and protocol implementation (UART, I2C, SPI, Modbus, TCP/IP). Experienced in developing high-availability failover mechanisms, industrial IoT gateways, and multi-threaded C applications for critical applications.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Data Structures, Bare-Metal Programming
- **Operating Systems & RTOS:** Embedded Linux, FreeRTOS, Linux Kernel, Linux System Programming
- **Protocols & Interfaces:** UART, I2C, SPI, USB, Modbus RTU/TCP, MQTT, TCP/IP
- **BSP & Build Systems:** Board Bring-up, Bootloader (U-Boot) Porting, Yocto Project, Linux Kernel Bring-up
- **Device Drivers:** I2C, SPI, UART, USB Driver Development & Character Device Drivers
- **Microcontrollers & SoCs:** AM335x, ESP32, LPC2148, STM32, Arduino Uno
- **Tools & Debugging:** GDB, Oscilloscopes, Logic Analyzers, Git, Makefile, OpenOCD

PROFESSIONAL EXPERIENCE

Kellton Tech | ***Embedded Developer***

April 2024 – Present | *Hyderabad, India*

- **Linux Kernel & Device Driver Development:** Developed and integrated custom Linux device drivers for I2C and SPI peripherals on AM335x platforms.
- **BSP & Board Bring-up:** Executed board bring-up procedures, custom bootloader (U-Boot) porting, and kernel configuration using Yocto layers.
- **Industrial IoT Gateway Deployment:** Installed, configured, and commissioned robust industrial IoT gateway panels for field operations (Oil India site), enabling real-time sensor data streaming over Modbus and TCP/IP.
- **Multi-Threaded C Applications:** Engineered high-performance, multi-threaded C applications for gateway communications using POSIX threads, IPC mechanisms, and socket programming.
- **System Stability & Drivers:** Implemented hardware Watchdog timers and low-level diagnostic routines to ensure maximum uptime in harsh field environments.

Vector India | ***Trainee – Advanced Embedded Systems***

August 2023 – February 2024

- Developed bare-metal drivers and firmware in C for ARM-based microcontrollers (LPC2148/STM32) handling UART, SPI, and I2C peripherals.
- Configured FreeRTOS tasks, queues, semaphores, and mutexes for reliable real-time task scheduling.

KEY PROJECTS

Redundant Controller Failover System (ESP32 / FreeRTOS)

- Designed a dual-controller hardware failover architecture using FreeRTOS on ESP32.
- Configured a custom pulse-based heartbeat (100ms interval, 500ms debounce threshold) over Bluetooth and established inter-module synchronization via UART to handle seamless failover without system downtime.

Smart Gate Keeping System (Automated Surveillance)

- Built an automated vehicle monitoring pipeline integrating Embedded Linux modules and computer vision models.
- Optimized hardware resource usage and peripheral communication for real-time edge processing.

Face Mask Detection (ESP32 Edge AI)

- An ESP32-based Edge AI system that detects with mask, without mask, and incorrectly worn mask from images captured through a web interface.
- The system uses a lightweight INT8 TensorFlow Lite Micro model for on-device inference, with Wi-Fi and an HTTP/HTTPS server for real-time predictions.

EDUCATION

B.Tech in Electronics and Communication Engineering

Indian Institute of Information Technology Design and Manufacturing (IIITDM), Kurnool (2019 – 2023)